```
import numpy as np
def cosine_similaity(a,b):
  return np.dot(a,b) / (np.linalg.norm(a) * np.linalg.norm(b))
good = np.array([0.82, 0.41, 0.10])
great = np.array([0.79,0.38, 0.12])
bad = np.array([-0.71, 0.12, -0.05])
print(f'good vs great: {cosine_similaity(good, great):.3f}')
print(f'good vs bad: {cosine_similaity(good, bad):.3f}')
print(f'great vs bad : {cosine_similaity(great, bad):.3f}')
```

```
good vs great: 1.000
good vs bad: -0.808
great vs bad : -0.816
```

```
!pip install gensim
from gensim.models import Word2Vec

sentences = [
'the king ruled the kingdom wisely',
'the queen ruled the land gracefully',
'the prince will become king one day',
'a man walked into the palace',
'a woman walked into the palace',
'the dog barked loudly in the garden',
'the puppy played happily in the garden',
]

tokenised = [s.split() for s in sentences]
model = Word2Vec(sentences=tokenised, vector_size=50,
window=3, min_count=1, sg=1, epochs=200)
print(model.wv.most_similar('king', topn=3))
result = model.wv.most_similar(
positive=['king', 'woman'],
negative=['man'],
topn=1
)
print('king - man + woman =', result[0][0])
```

```
Collecting gensim
  Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8.4 kB)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.6.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.2.1)
Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 27.9/27.9 MB 30.0 MB/s eta 0:00:00
Installing collected packages: gensim
Successfully installed gensim-4.4.0
[('kingdom', 0.40141454339027405), ('played', 0.38476690649986267), ('in', 0.38395753502845764)]
king - man + woman = ruled
```

```
import sys
!{sys.executable} -m pip install gensim
# Step 0: Download GloVe vectors
# https://nlp.stanford.edu/data/glove.6B.zip (about 800 MB)
# Extract and you get: glove.6B.50d.txt, glove.6B.100d.txt, etc.
!wget --no-check-certificate https://nlp.stanford.edu/data/glove.6B.zip
!unzip -q glove.6B.zip

# Step 1: Convert GloVe format to Word2Vec format (one-time step)
from gensim.scripts.glove2word2vec import glove2word2vec
glove2word2vec('glove.6B.100d.txt', 'glove.6B.100d.w2v.txt')
# Step 2: Load the converted file
from gensim.models import KeyedVectors
glove = KeyedVectors.load_word2vec_format(
'glove.6B.100d.w2v.txt', binary=False
) # Takes ~30 seconds; 400,000 words x 100 dimensions
# Step 3: Use it!
# --- Find top-5 most similar words ---
print(glove.most_similar('bollywood', topn=5))
# [('cinema', 0.82), ('films', 0.79), ('actors', 0.77), ...]
# --- Test analogy ---
res = glove.most_similar(positive=['paris','AP'], negative=['france'],
topn=1)
```

```
print('paris - france + india =', res[0][0]) # hopefully 'delhi'!
# --- Cosine similarity between two words ---
print(glove.similarity('dog', 'puppy')) # ~0.81
print(glove.similarity('cat', 'cricket')) # ~0.12 (unrelated)
```

```
Requirement already satisfied: gensim in /usr/local/lib/python3.12/dist-packages (4.4.0)
Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) (2.0.2)
Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) (1.16.3)
Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) (7.6.1)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1->gensim) (2.2.1)
--2026-06-11 05:49:14--  https://nlp.stanford.edu/data/glove.6B.zip
Resolving nlp.stanford.edu (nlp.stanford.edu)... 171.64.67.140
Connecting to nlp.stanford.edu (nlp.stanford.edu)|171.64.67.140|:443... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip [following]
--2026-06-11 05:49:15--  https://downloads.cs.stanford.edu/nlp/data/glove.6B.zip
Resolving downloads.cs.stanford.edu (downloads.cs.stanford.edu)... 171.64.64.22
Connecting to downloads.cs.stanford.edu (downloads.cs.stanford.edu)|171.64.64.22|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 862182613 (822M) [application/zip]
Saving to: ‘glove.6B.zip.1’

glove.6B.zip.1      100%[===================>] 822.24M  5.07MB/s    in 3m 1s

2026-06-11 05:52:17 (4.53 MB/s) - ‘glove.6B.zip.1’ saved [862182613/862182613]

replace glove.6B.50d.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: /tmp/ipykernel_6373/2097256229.py:11: DeprecationWarning: Call t
  glove2word2vec('glove.6B.100d.txt', 'glove.6B.100d.w2v.txt')
---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
/tmp/ipykernel_6373/2097256229.py in <cell line: 0>()
      9 # Step 1: Convert GloVe format to Word2Vec format (one-time step)
     10 from gensim.scripts.glove2word2vec import glove2word2vec
---> 11 glove2word2vec('glove.6B.100d.txt', 'glove.6B.100d.w2v.txt')
     12 # Step 2: Load the converted file
     13 from gensim.models import KeyedVectors

                                5 frames
/usr/local/lib/python3.12/dist-packages/gensim/models/keyedvectors.py in _word2vec_line_to_vector(line, datatype,
unicode_errors, encoding)
   1980 def _word2vec_line_to_vector(line, datatype, unicode_errors, encoding):
   1981     parts = utils.to_unicode(line.rstrip(), encoding=encoding, errors=unicode_errors).split(" ")
-> 1982     word, weights = parts[0], [datatype(x).item() for x in parts[1:]]
   1983     return word, weights
   1984 

KeyboardInterrupt: 
```

```
import gensim.downloader as api
glove_sm = api.load('glove-wiki-gigaword-50')
print(glove_sm.most_similar('bollywood', topn=3))
```

```
[('malayalam', 0.7608863115310669), ('films', 0.7456727623939514), ('hindi', 0.7311806678771973)]
```